

When Not to Generate: How AI Systems Quote Scripture, and What Authoritative Quotation Requires

Alex Chao
Fide AI
alex@fideai.org

July 2026

Abstract

People increasingly ask AI assistants for Scripture—in sermon preparation, teaching, and pastoral conversation. These are not ordinary information requests: someone asking for a passage wants particular words, in a particular translation, from a source they could have opened themselves. A reply that blends two translations, drops a verse, or paraphrases inside quotation marks satisfies none of that while appearing to satisfy all of it, and the person who asked is the one least able to check. Existing evaluation does not see this failure. A response can be factually accurate, properly cited, and semantically faithful, and still be the wrong text.

When the words themselves are the answer, who should be responsible for them? We route matched Scripture requests through four designs—the model quotes from memory; the passage is placed in front of it to copy; it is given a lookup tool; or it only names the passage and non-generative code inserts the text. Locked in advance, the study covers 8,640 requests across six model families, three open English editions, and twenty passages. Quoting from memory produced the requested edition exactly 25.00% of the time; supplying the text raised that to 93.61%, a lookup tool to 80.09%, and deterministic insertion to 83.94%. The central finding is that connecting a source did not remove failure but moved it onto whatever the model still decided. Models called the tool in 95.00% of cases yet asked it for the right passage in only 84.17%. Deterministic insertion was near-perfect once the model named the right passage—1,813 of 1,815—and powerless when it did not. Exact quotation is not one skill but a chain, and each link fails on its own terms. This measures textual delivery, not whether a passage was well chosen, rightly interpreted, or pastorally apt.

1 Introduction

Consider three requests a person might make of an AI assistant. “Give me Ephesians 2:8–9 in the Berean Standard Bible for Sunday’s slide.” “Quote the passage where Jesus compares his being lifted up to Moses’ serpent in the wilderness.” “Send me Revelation 21:1–4 to read at a funeral.” Each asks for exact words, but each puts a different decision in the system’s hands. The first names the passage and the edition, leaving only faithful reproduction. The second names neither, so the passage must be identified before it can be quoted. The third is long enough that truncation, not misidentification, is the likely failure. An assistant that answers all three the same way—by generating fluent text that carries the right reference—fails each of them differently, and the final prose does not reveal which failure occurred.

That opacity is the practical problem. The meaning may survive a blended or shortened rendering; the quotation does not. And the person who asked is poorly placed to notice, because checking the answer would require already having the text they asked for. For requests like these, wording is not merely a vehicle for the answer. Wording is part of the answer.

These requests are not hypothetical, and the settings are not peripheral. In a December 2025 survey of U.S. Protestant pastors, 87% reported using AI in ministry, 36% for biblical or theological research, and 24% for writing or editing sermons (Barna Group, 2026). A separate survey of 594 church leaders found comparable adoption alongside near-absent governance: most reported frequent use, while 73% reported no AI policy of any kind (Exponential AI NEXT and ChurchTechToday, 2025). Which English translation a person reads is itself a durable, measurable feature of American religious life rather than an incidental preference (Goff et al., 2017), so substituting one edition for another is not a cosmetic error.

The stakes are practical and specific. A congregation reads responsively from a printed edition while a screen displays different wording. A student memorizes a verse in one translation and is later corrected against another. A teacher builds an argument on a clause that the quoted edition does not contain. Someone in distress receives a passage with its final line missing. In each case the response is fluent, carries the right reference, and is wrong in the way that matters. Readers reasonably treat quotation marks and a verse reference as a claim about a source of record; when that claim is unreliable and difficult to detect, the cost falls on the reader rather than on the system.

Practitioners have noticed. The chief executive of a major digital Bible platform has publicly estimated that leading models misquote Scripture between 15% and 60% of the time, and has cited that estimate as a reason not to ship a Scripture-answering assistant (Christian Daily International, 2026). The estimate is plausible and consequential, but no measurement protocol accompanies it: the systems, passages, editions, and criterion for “misquote” are unstated, so it can be neither checked nor acted on with confidence. Claims of that importance deserve a specified, reproducible measurement. Providing one, for a bounded set of English editions and conditions, is the purpose of this paper.

We use *authoritative quotation* to mean delivering the requested span, in the requested edition, from an identified source of record—a specific, versioned text designated as authoritative for that request, rather than an abstract notion of authority. A response may be relevant, factually close, or theologically unobjectionable while still failing that narrower task. It may silently substitute an edition, blend familiar formulations, omit a verse, expand beyond the requested span, or present a paraphrase as a quotation. This is a stricter requirement than factuality, attribution, or groundedness: a response can be factually supported and properly cited while still failing an exact quotation request.

Neither model scale nor retrieval resolves the problem on its own. Language models are built to generate plausible continuations; parametric memory can sometimes reproduce a familiar source verbatim, but it is not a source-of-record interface. Retrieval can make authoritative text available, and a generative layer can still alter it. A model holding a tool can bypass the tool or request the wrong span. A deterministic renderer can preserve source bytes exactly, but only downstream of a correct decision about source, edition, and boundaries. Return to the serpent request above and the point sharpens: a model answering from memory may blend familiar translations; a model given the source may shorten it; a tool-enabled model may retrieve the neighboring verse; and a deterministic renderer may reproduce the source perfectly after the model selected the wrong range. The four designs fail at four different places, and every one of them produces credible prose. Authoritative quotation is a property of the whole source-delivery chain, not a model knowledge property.

We therefore hold the user’s request fixed and vary which component owns the words. Matched

requests were routed through four architectures: generation from model memory, reproduction of source text supplied in context, retrieval through an authorized `get_passage` tool, and model selection of a structured reference followed by deterministic rendering. The prospectively specified and internally hash-locked design contains 8,640 observations across six model families, three open English editions, twenty passages, two prompt forms, and three repetitions. This is a matched crossed comparison rather than a benchmark or leaderboard, and the primary endpoint is deliberately demanding: the user-visible output must equal the requested text exactly, with condition-specific evidence that the declared architecture was actually followed. Scripture makes this comparison unusually tractable—verse references supply a conventional addressing scheme, multiple English translations make edition identity observable rather than assumed, span length can be varied systematically, and indirect prompts force semantic identification before rendering.

The results are stark and, we think, more interesting than a ranking. Native generation from model memory met the strict endpoint in 25.00% of observations, against 93.61% for source-supplied quotation, 80.09% for tool-mediated retrieval, and 83.94% for deterministic rendering. Because the endpoint requires the *requested* edition, much of the native gap is edition substitution rather than general unfamiliarity: native exactness ranged from 41.53% for the Berean Standard Bible to 5.00% for the Literal Standard Version. The central finding is what happened after a source was connected. Failure did not disappear; it migrated toward whichever decision the architecture still left to the model. A source tool was invoked in 95.00% of tool observations but used for the requested lookup in only 84.17%. Deterministic rendering was exact in 1,813 of 1,815 cases once the model had selected the correct reference—but the model selected correctly in only 84.03% of observations, and no renderer can repair a wrong selection. Each architecture made one stage reliable and left another exposed.

What transfers from this is the structure, not the numbers. The same shape appears wherever a source of record exists: a dosing instruction, a statutory subsection, an indemnity clause, and a safety standard each have an edition, a boundary, and wording that a paraphrase does not preserve, and each can fail at selection, access, rendering, or final output. The decomposition and evaluation design developed here apply to those settings directly. The measured rates do not—every number in this paper is about English Bible quotation, and another domain needs its own sources, scenarios, and expertise. Many faith traditions likewise designate specific textual forms as authoritative and treat exact wording as consequential, but the unit of authority differs—a source-language consonantal text, a recited form, a received translation, a commentary tradition—and those differences would change the study design, not only the results. We report on English Christian Bible editions and make no claim about any other tradition’s texts. Textual fidelity is also not theological fidelity: the study evaluates whether a system delivered the requested source faithfully, not whether the passage was well chosen, rightly interpreted, or pastorally appropriate.

Contributions. The paper makes three contributions. First, it gives a rigorous empirical account of Scripture quotation across twenty passages, three open English editions, six model families, and four system architectures. Second, it defines a *source-delivery chain* that separates source selection, authorized access, exact rendering, and final-output integrity, and shows that quotation failures are only interpretable when attributed to a stage. Third, it releases an open evaluation methodology, deidentified derived scores, and reproducible analyses while keeping private execution code, raw model traces, and rights-constrained text outside the public artifact. The study answers a public call for research on sacred-text fidelity (Fide AI, 2026), and the repository records which parts of that call remain open. Three questions organize the study: how architecture-adherent exactness differs across the four architectures; at which stage failures occur; and how prompt form, passage

length, edition, model family, and repeated sampling interact with those failures.

2 Authoritative Quotation as a Systems Problem

2.1 Quotation succeeds only when the whole chain succeeds

A successful quotation needs four things to go right. The system must identify what text the user means, obtain it through the declared path, preserve its wording, and deliver that wording intact. Failure at any one stage defeats the quotation request even if every other stage succeeds.

Formally, let a request specify, explicitly or implicitly, a source work W , an edition E , and a span B . Let $T(W, E, B)$ be the authoritative text returned by the source of record. We separate four system properties:

- S : the intended work, edition, and span are selected correctly;
- A : the declared source-access method is followed and the requested source is available;
- R : the selected source text is rendered without alteration; and
- I : the user-visible final output preserves that rendering without truncation, leakage, or extraneous text.

The strict endpoint is conjunctive:

$$\text{E2E-EXACT} = S \wedge A \wedge R \wedge I. \quad (1)$$

The operational evidence for each term depends on architecture. In native generation there is no external access event to verify; exact final output is the decisive behavior. In a tool condition, access requires an observed call for the requested span rather than merely the presence of a tool definition. In a deterministic condition, exact rendering is meaningful only after a parseable, correct reference has reached the renderer.

This decomposition prevents a common category error. A renderer may provide a strong guarantee about R while the overall system still fails because S is false. Conversely, a model may identify the correct passage while a generative surface layer changes the words. An aggregate score is useful, but stage-level evidence is required to explain it.

2.2 Why factuality and citation are not enough

A quotation can be factually supported yet textually wrong. A citation can point to the correct document while the quoted span omits a limiting clause. A semantic similarity score can treat two Bible translations as nearly equivalent even when the user requested one exact edition. Authoritative quotation is therefore stricter than factuality, attribution, or general groundedness.

The requirement is also narrower. The evaluation does not ask whether the source itself is true, wise, clinically appropriate, legally controlling, or theologically interpreted correctly. It asks whether a system delivered the requested source faithfully under its declared architecture.

2.3 Scripture as the study domain

Scripture warrants study because AI systems increasingly mediate access to texts that Christian communities receive, interpret, and transmit as authoritative, and because the domain makes the quotation problem measurable. A reference such as “John 3:16” identifies a conventional verse

span but not one English wording, so edition identity is observable rather than assumed. An event-style request adds a prior selection problem: the system must identify the intended span before it can quote it. Few domains combine a stable addressing convention, multiple simultaneously authoritative editions, and a community that treats exact wording as consequential.

Working here requires textual and theological restraint. The English editions studied sit within a complex manuscript and translation history in both the Hebrew Bible and New Testament (Tov, 2012; Metzger and Ehrman, 2005), and translation requires linguistic and interpretive decisions (Nida and Taber, 1969). The three editions differ not only in translation philosophy but in underlying text-critical basis, so “edition” is partly a claim about source-language witnesses rather than purely a matter of English wording. The twenty targets were not selected to probe variant units, and a study designed around textual variants would be a distinct and worthwhile contribution. Exact reproduction of a selected English edition is therefore not fidelity to the Hebrew, Aramaic, or Greek source texts, and no result here establishes that one English translation is doctrinally or linguistically superior. The twenty targets all fall within books shared by the major Christian canons; the study does not adjudicate canonical boundaries, and the three editions carry the Protestant canon.

For churches and Christian institutions, the practical implication is correspondingly bounded. A system that quotes the requested edition exactly removes one avoidable textual failure. It does not replace biblical scholarship, ecclesial authority, pastoral judgment, or attention to literary and canonical context. Exact quotation is a prerequisite for some forms of responsible use, not a complete account of responsible use.

3 Related Work

Retrieval-augmented generation combines parametric sequence models with retrieved evidence (Lewis et al., 2020). Later work examines when parametric memory is insufficient and non-parametric memory should be preferred (Mallen et al., 2023), while RAGAs evaluates retrieval and generation quality within retrieval-augmented systems (Es et al., 2024). Tool-use research asks whether models can decide when and how to invoke external APIs (Schick et al., 2023; Ross et al., 2025), and grammar-constrained decoding offers one way to enforce structured interfaces (Geng et al., 2023). Authoritative quotation joins these concerns: the system must obtain a declared span, demonstrate that the intended access path was used, and preserve the result.

Fine-grained factuality work decomposes long-form generation into checkable claims (Min et al., 2023), while citation-oriented systems evaluate whether generated claims are accompanied by supporting sources (Gao et al., 2023). Surveys document broad forms of unsupported and unfaithful neural generation (Ji et al., 2023; Huang et al., 2025). Our task is stricter in wording and narrower in scope: a response can be factually supported and properly cited while still failing an exact quotation request.

CopyBench measures literal and non-literal reproduction of copyrighted fiction books (Chen et al., 2024), while Quote-Tuning increases verifiability by aligning models to quote verbatim strings found in trusted pretraining corpora (Zhang et al., 2025). The present study differs in object and architecture: the requested edition and span are specified as source-of-record requirements, and authorized retrieval or deterministic rendering can replace parametric reproduction. These works are complementary and bound the novelty claim; this paper does not claim to introduce verbatim-copy measurement or quoting as a general research problem. Training-data extraction and memorization studies likewise show that models can reproduce memorized content under some conditions (Carlini et al., 2021, 2023). Memorization does not make parametric recall a stable or

authorized source interface. Copyright policy work, including the United States Copyright Office’s generative-AI training report, provides further context for making source access and redistribution assumptions explicit ([United States Copyright Office, 2025](#)).

Holistic and behavioral evaluation emphasizes scenario coverage, multiple metrics, and tests of specific capabilities rather than one undifferentiated score ([Liang et al., 2023](#); [Ribeiro et al., 2020](#)). We adopt that orientation without turning the study into a ranking exercise. The contribution is the source-delivery chain: a matched evaluation of which component selected, obtained, rendered, and preserved the requested source.

4 Study Design

The experiment holds the request panel fixed and changes the route by which a quotation reaches the user. This makes architecture, rather than passage choice or model mix, the primary comparison.

4.1 Prospective specification and execution boundary

The primary study was prospectively specified and internally hash-locked before confirmatory execution began. This was not a public registry deposit. The lock fixed the target pack, source registry, prompt families, model routes, conditions, repetitions, scoring rules, denominator policy, and bootstrap procedure. Integration and calibration runs were excluded from confirmatory estimates. Post-lock administrative changes removed administrative stopping ceilings and divided execution into durable blocks; they did not change a prompt, source, model, observation, score, or analysis.

The public replication package records the lock time, input digests, shared implementation revision, and post-lock deviations. It also releases the exact executed system and caller templates, target substitutions, functional tool schema, and sampling/turn policy. Authoritative passage text is omitted from the prompt artifact, but its release-safe digest is retained where applicable.

The primary phase contained:

$$20 \text{ targets} \times 2 \text{ prompt families} \times 4 \text{ conditions} \times 3 \text{ editions} \times 6 \text{ model routes} \times 3 \text{ epochs} = 8,640. \quad (2)$$

The unit of analysis is one target–prompt–condition–edition–model–epoch observation. Conditions are paired within all other factors.

4.2 Passage targets

The target set is a stratified purposive panel, not a probability sample of the Bible or of user requests. It contains five targets in each of four strata:

1. explicit single verses;
2. explicit short multi-verse passages;
3. longer passages intended to expose truncation, span, or guardrail behavior; and
4. event-style or semantic reference requests.

Targets include narrative, poetry, wisdom, prophetic, Gospel, and epistolary material. Two known intertextual or adjacent-passage relationships were annotated rather than treated as fully independent evidence: the Shema and great-commandment pair, and adjacent passages in John 3.

Each target had an explicit-reference prompt and an edition-neutral contextual description. The contextual descriptions did not name the book, chapter, or verse and were designed to avoid distinctive wording from any tested edition. An initial review identified six descriptions requiring revision. Those six underwent a structurally blinded focused reapproval and produced six exact reference identifications. The original twenty-item review was not fully blinded, so only the focused six-item result is described as blinded evidence.

4.3 Source editions

The full four-condition comparison used three open English editions:

- Berean Standard Bible (BSB), a modern public-domain edition ([Berean Bible, 2023](#));
- World English Bible Updated (WEBU), a public-domain translation ([eBible.org, 2024](#)); and
- Literal Standard Version (LSV, 2020), released under CC BY-SA ([Covenant Press, 2020](#)).

These editions differ in translation philosophy and in underlying text-critical basis, so a requested edition specifies more than a style of English.

A source of record is a versioned artifact, so edition identity is disclosed rather than left to the edition name. The public package includes a release-safe source-edition record giving, for each edition, the provider and provider-side source identifier, the edition designation, the rights basis, and a normalized-text SHA-256 digest for each of the twenty targets. Ground truth was verified at 2026-07-26T20:38:10Z by two uncached retrievals through the pinned shared runtime with matching digests; this establishes retrieval stability, not independence of the upstream provider. The same stored text then served as ground truth for every observation. Because the digests are computed over whitespace-collapsed UTF-8 text, a replicator can hash their own copy of a passage and confirm they hold the same edition text without either party transmitting it. The record is derived from the source registry fixed by the prospective lock and carries that artifact’s digest, so the disclosure can be checked against a commitment made before confirmatory execution.

Authoritative source text was retained privately for scoring. The public result package contains aggregate metrics and no LSV passage text. Four restricted editions were present in the locked registry but excluded from the primary phase pending source and source-to-model authorization; they are listed in the source-edition record with that status.

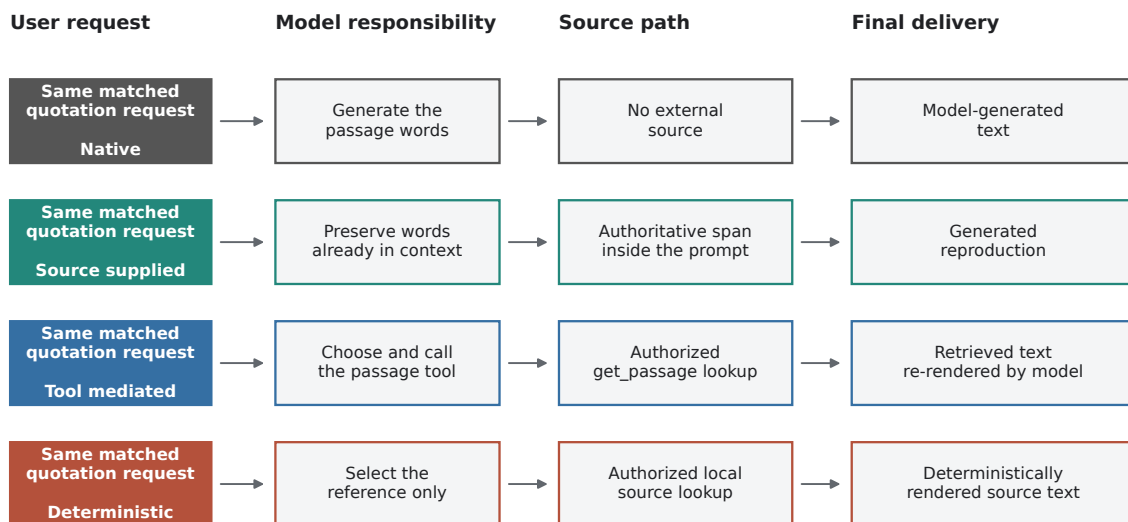
4.4 Model routes and sampling

The primary panel used one route from each of six model families: GPT-5.6 Sol, Claude Sonnet 5, Gemini 3.5 Flash, DeepSeek V4 Pro, Kimi K3, and GLM-5.2. These labels identify the endpoints evaluated on July 26–27, 2026; they are not claims about later versions or general vendor performance.

Requests used OpenRouter-compatible routing with one upstream provider pinned, fallbacks disabled, required parameter support, and data-collecting routes denied. Provider-default sampling was used with temperature omitted. Every cell was observed in three epochs. Semantic retries were zero. Bounded transport retries were allowed, and terminal provider errors remained non-deliveries in the primary denominator.

4.5 Architecture conditions

The four conditions give the model progressively different responsibilities. In the first, the model must supply the words itself. In the second, the words are already present and the model must



The user request is held fixed. What changes is which component owns selection, source access, and the final words.

Figure 1: The four evaluated paths through the source-delivery chain. The caller request is matched across conditions; what changes is which component owns the source words and the decisions that precede them.

preserve them. In the third, the model must obtain the words through a tool and then return them. In the fourth, the model selects only the reference; a deterministic layer owns the words. The caller asks the same question in every matched cell, so the comparison changes the delivery path rather than the user’s request (Figure 1).

Native parametric quotation. The model received no source text and no tool. This condition represents the common behavior of asking a general assistant to quote from memory.

Source-supplied quotation. The authoritative span was placed in the model context and the model was asked to return it unchanged. This approximates a simple retrieval or source-injection architecture while isolating whether the generative layer preserves available text.

Tool-mediated retrieval. The model received an authorized `get_passage` tool and was instructed to use it. The trace recorded whether the tool was invoked and whether it retrieved the requested span. Emitting text that resembled a tool call without an executable call did not count as tool use.

Structured selection and deterministic rendering. The model was instructed to return exactly one placeholder of the form `<quote>{{QUOTE:<reference>}}</quote>`. A local deterministic layer parsed the reference, fetched the selected edition and span, and replaced the placeholder. The primary score used the strict executed grammar. A prospectively specified analysis-only tolerant parser separately assessed whether one unambiguous expected reference could be recovered from malformed formatting; it never changed the observed output or primary score.

4.6 Outcomes and estimands

Primary and common outcomes. The locked primary endpoint, E2E-EXACT, required strict final-output equality and all condition-specific selection, lookup, replacement, and method-adherence checks. We call it *architecture-adherent exactness* throughout: the gates differ by condition because the declared architectures differ. The common cross-condition text outcome is final-output exactness, which asks only whether the complete user-visible output equals the requested text. Both are reported so that a process failure is not mistaken for a textual mismatch, or vice versa. No weighted composite score was used. Terminal errors counted as failures in the intention-to-observe denominator.

We also report:

- exact and declared-normalized text match;
- answer presence and span coverage;
- strict and tolerant reference selection;
- tool invocation, requested lookup, and bypass;
- placeholder parsing and deterministic replacement; and
- final-output exactness and extraneous text.

Normalization covered limited surface variation and was never used to erase a translation or edition difference. The strict endpoint remained byte-sensitive to the protocol’s expected final text after declared wrapper handling.

Estimands and uncertainty. Condition rates use all 2,160 scheduled observations per condition and are accompanied by raw counts. The prospectively specified primary contrast is the paired risk difference between deterministic rendering and native generation. We also report paired source-supplied and tool-mediated contrasts. Paired intervals come from a 10,000-resample target-cluster bootstrap with seed 5601. Clustering by target acknowledges repeated observations of the same passage while preserving the crossed cells inside each resample.

Model, prompt, edition, and passage-stratum analyses are descriptive interactions. The study was not designed to rank vendors, estimate population prevalence across Scripture, or infer training-set membership. Model routes and editions are a fixed evaluation panel; the bootstrap does not support generalization to untested routes or editions.

We additionally report target-level rate distributions and a post-hoc linear probability sensitivity model with condition, route, edition, prompt-family, passage-stratum, and epoch fixed effects and target-cluster-robust standard errors using a t_{19} reference distribution. This model was not part of the prospective lock and is used only to test whether the primary fixed-panel conclusions depend on the unadjusted presentation. It does not convert the purposive target panel into a probability sample.

5 Results

Three findings organize the results. First, every source-connected condition delivered exact text substantially more often than generation from model memory. Second, connecting a source did not remove failure; it moved failure to a different stage of the source-delivery chain. Third, passage length and the need to infer a reference exposed weaknesses that pooled condition rates alone

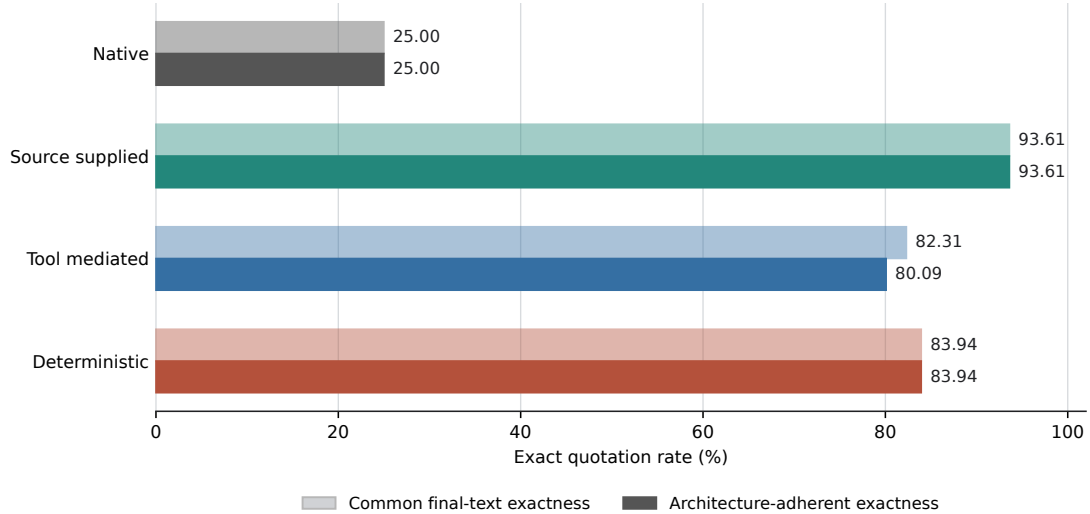


Figure 2: Common final-text exactness and architecture-adherent exactness by condition. The measures differ only for tool-mediated retrieval, where some exact final answers did not satisfy the declared tool-use gates.

Condition	Final exact	E2E-EXACT exact	N	E2E-EXACT rate
Native parametric quotation	540	540	2,160	25.00%
Source-supplied quotation	2,022	2,022	2,160	93.61%
Tool-mediated retrieval	1,778	1,730	2,160	80.09%
Deterministic rendering	1,813	1,813	2,160	83.94%

Table 1: Common final-text exactness and architecture-adherent exactness. All scheduled observations, including terminal provider errors, remain in the denominator.

would miss. We begin with execution integrity and the primary comparison, then trace those failures through the chain.

5.1 The scheduled design was retained despite provider errors

All 48 route–method–prompt–family blocks completed, yielding all 8,640 scheduled observations and unique trial and execution-request identities. Thirty-three observations ended in terminal provider errors: 23 on the DeepSeek route and 10 on the Kimi route. Thirty-two errors occurred in native generation and one in tool-mediated retrieval. No semantic replacement sample was run.

Response-level provider and response-ID metadata were present for 8,594 observations. The 46 gaps were confined to DeepSeek or Kimi native-generation observations; 32 were terminal errors and 14 completed. The one remaining terminal error occurred in tool-mediated retrieval and retained its provider metadata. Requested routes still had one upstream pinned and fallbacks disabled. We retain this as disclosed provenance missingness rather than excluding the observations.

Treatment vs. native	Paired risk difference	Cluster bootstrap 95% interval
Source supplied	+68.61 percentage points	[+61.71, +75.60]
Tool mediated	+55.09 percentage points	[+46.48, +63.47]
Deterministic rendering	+58.94 percentage points	[+51.16, +66.67]

Table 2: Paired effects on architecture-adherent exactness. Complete pairs: 2,160 for each contrast.

Condition	Target median	IQR	Minimum	Maximum
Native parametric	18.98%	[10.65, 39.58]	0.00%	56.48%
Source supplied	98.61%	[92.82, 100.00]	55.56%	100.00%
Tool mediated	88.43%	[72.45, 94.44]	34.26%	96.30%
Deterministic rendering	89.35%	[83.33, 92.59]	46.30%	96.30%

Table 3: Distribution of architecture-adherent exactness across the twenty targets. Each target rate pools the balanced route, edition, prompt, and epoch panel ($N = 108$ per condition and target).

5.2 Source-connected systems quoted far more exactly than generation from memory

Native parametric quotation produced an exact final quotation in one quarter of observations (Figure 2; Table 1); in other words, three out of four requests failed the strict endpoint when the model had to supply the passage from memory. Every source-connected intervention substantially improved on that baseline. Source-supplied quotation had the highest pooled rate, followed by deterministic rendering and tool-mediated retrieval. That ordering is useful descriptively, but it does not yet explain what each architecture made reliable.

The prospectively specified deterministic-versus-native risk difference was +58.94 percentage points (Table 2). The source-supplied contrast was larger at +68.61 percentage points, and the tool-mediated contrast was +55.09 percentage points. The intervals describe variation across the twenty target clusters, not a probability sample of all possible Bible passages or user requests.

The target distributions show that the pooled effects were not driven by one or two passages (Table 3), but they also expose substantial target heterogeneity. In the post-hoc cluster-robust sensitivity model, adjusted risk differences were +68.61 percentage points for source supplied (t_{19} interval [+60.83, +76.39]), +55.09 percentage points for tool mediated ([+45.68, +64.50]), and +58.94 percentage points for deterministic rendering ([+50.36, +67.51]). These estimates are nearly identical to the paired fixed-panel contrasts and are interpreted as sensitivity results, not new confirmatory claims.

5.3 Each architecture left a different model decision exposed

The aggregate rates become more informative when read as a sequence of gates (Table 4). Source-supplied quotation removed the first two gates: the model did not have to identify or retrieve the passage because the authoritative span was already in context. Even so, 138 final outputs were inexact. Making authoritative text available therefore approached, but did not create, deterministic copying.

The tool condition reintroduced source access as a model decision. A tool invocation was observed in 2,052 of 2,160 observations (95.00%), but the requested retrieval was used in only 1,818 (84.17%). The remaining trace-level failures included 107 bypasses and 234 calls for a span other than the requested span, plus one terminal error. Among observations satisfying tool method

Condition	Answered	Exact text	Method adherence	Final exact	E2E-EXACT
Native	98.29%	25.14%	–	25.00%	25.00%
Source supplied	100.00%	93.66%	100.00%	93.61%	93.61%
Tool mediated	99.81%	82.31%	84.17%	82.31%	80.09%
Deterministic rendering	100.00%	83.98%	92.55%	83.94%	83.94%

Table 4: Selected component rates. “Method adherence” is condition-specific and should not be compared as a general model capability.

Condition	Explicit reference	Contextual description
Native	27.41%	22.59%
Source supplied	93.52%	93.70%
Tool mediated	90.19%	70.00%
Deterministic rendering	85.56%	82.31%

Table 5: Architecture-adherent exactness by caller prompt family ($N = 1,080$ per cell).

adherence, 1,730 of 1,818 (95.16%) achieved architecture-adherent exactness. The tool itself was not the guarantee; correct use of the tool was.

The deterministic condition moved responsibility again. Strict reference selection was correct in 1,815 of 2,160 observations (84.03%). Among those correct selections, 1,813 produced an exact final quotation (99.89%). Once the correct reference crossed the interface, the renderer behaved almost deterministically. The dominant uncertainty was upstream: whether the model selected the right passage.

Formatting and selection were related but not identical. The outer placeholder was recognized in 2,159 observations, while 161 observations carried a syntax-related failure tag, usually because the model added an edition label to the reference payload. The analysis-only tolerant parser identified the expected reference in 1,954 observations (90.46%), including 158 of those 161 syntax-tagged cases. This does not change the strict 83.94% primary result. It shows that interface conformance and semantic reference selection are different questions.

5.4 Contextual requests mainly challenged reference identification

Contextual descriptions had little pooled effect on source-supplied reproduction, where the authoritative span was already present. Their largest effect appeared in tool-mediated retrieval: exactness fell by 20.19 percentage points relative to explicit references. Deterministic rendering fell by 3.25 percentage points, consistent with the model having to identify a span before the renderer could act.

5.5 Long passages remained difficult even with source access

Native exactness varied sharply by edition (Table 6), reaching 41.53% for BSB and 5.00% for LSV. Source-connected conditions were less edition-sensitive. These results establish an interaction with requested edition in this panel. They do not identify its cause and cannot be used to infer undisclosed training-data frequency.

Long passages were the most difficult stratum under every architecture (Figure 3). Native exactness fell to 5.37%. Even when the source was supplied, long-passage exactness was 85.93%,

Condition	BSB	WEBU	LSV
Native	41.53%	28.47%	5.00%
Source supplied	94.44%	93.75%	92.64%
Tool mediated	80.97%	79.72%	79.58%
Deterministic rendering	87.92%	83.61%	80.28%

Table 6: Architecture-adherent exactness by requested edition ($N = 720$ per cell).

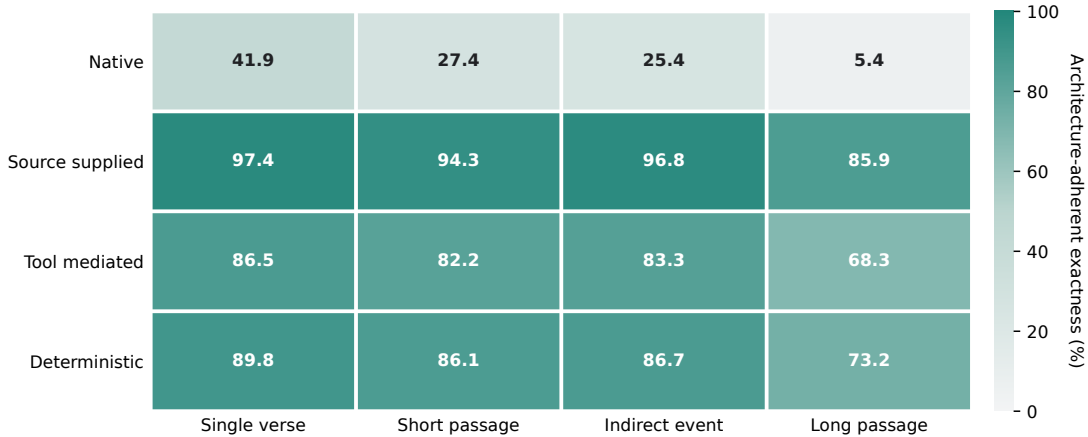


Figure 3: Architecture-adherent exactness by passage stratum ($N = 540$ per cell). “Indirect event” describes target construction; both prompt families were evaluated for every target.

indicating that rendering failures were not limited to source selection. Tool and deterministic conditions also lost accuracy through span selection before exact text could be delivered.

5.6 Effects varied by route but persisted across repetitions

Every route improved substantially over its own native condition, but the relative ordering of interventions differed (Appendix Table 7). This heterogeneity cautions against treating “RAG,” “tool use,” or “deterministic rendering” as model-independent properties.

Aggregate epoch rates were stable: 24.17–25.56% for native, 93.47–93.75% for source supplied, 79.72–80.69% for tool mediated, and 83.47–84.17% for deterministic rendering. Cell-level behavior was not perfectly stable. Of 720 matched cells per condition, all three epochs succeeded in 122 native, 650 source-supplied, 504 tool-mediated, and 537 deterministic cells. Repetition therefore added information beyond a single run.

5.7 Failure review located errors at different stages

After primary scores were frozen, we conducted a route-blinded qualitative review. Six failure strata were defined and five observations per stratum were selected by a deterministic SHA-256 ordering of trial identifiers. The review was descriptive and did not alter scores.

Native failures included near-exact surface differences, lexical drift toward another familiar rendering, incomplete spans, blended wording, and refusals. Source-supplied failures included both typography-only differences and substantive alteration or omission. Tool failures separated into bypass, tool-like text without an executable call, and authentic retrieval of the wrong boundaries. Deterministic selection failures frequently expanded an event to include narrative setup, narrowed

a requested range to its most salient verse, or appended an edition label that violated the strict interface.

These observations support mechanism-level interpretation, not subtype prevalence estimates. No raw response is included in the public manuscript.

5.8 Provider errors did not explain the main result

Complete-case sensitivity changed native exactness from 25.00% to 25.38% and tool exactness from 80.09% to 80.13%; the all-scheduled-observation analysis remains primary.

The run used 1,954,958 input tokens, 3,285,730 output tokens, and 2,772,266 reasoning tokens. The execution recorded 11,483 provider API attempts, including bounded transport retries, and 10,641 successful model responses. Provider-attempt counts are not equivalent to semantic turns.

6 Discussion

6.1 Quotation reliability moved through the source-delivery chain

The simplest reading of the primary table is that source-connected architectures outperformed generation from memory. That is true, but incomplete. The more useful finding is that each architecture made a different part of the journey reliable.

Source supply made identification and retrieval unnecessary, leaving preservation as the main risk. Tool mediation gave the model access to a source of record, but also gave it decisions about whether to call the tool and which span to request. Deterministic rendering made the words reliable after a successful handoff, while leaving reference selection exposed. The failures did not disappear as architecture changed; they migrated toward the remaining model-owned decisions.

The study therefore does not show that deterministic rendering is universally better than source-supplied quotation. It shows why guarantees must be assigned to stages. A renderer can guarantee the bytes it emits without guaranteeing that the model asked for the right source. A tool can make authoritative text available without guaranteeing that the model will use it. And a source placed in context can still be changed by the generative layer that presents it.

6.2 Implications for frontier AI systems

For frontier labs and application developers, the results suggest five design principles.

1. **Treat exact quotation as an intent class.** A system should distinguish requests for meaning from requests for exact source text.
2. **Make the source of record explicit.** Work, edition, span, authorization, and retrieval provenance should be represented in the system state rather than inferred only from generated prose.
3. **Separate selection from rendering.** Structured selection makes upstream uncertainty observable and permits a non-generative component to own final text.
4. **Verify tool behavior, not tool availability.** A tool definition does not establish invocation, correct lookup, or faithful rendering.
5. **Test the final user-visible buffer.** Exact source bytes can still be lost through truncation, wrappers, streaming errors, or later generative transformation.

These principles generalize naturally to statutes, standards, policy manuals, clinical instructions, contracts, and other materials where a source of record exists. Generalization of the measured effect sizes, however, requires domain-specific replication.

6.3 Implications for Christian institutions

The results give Christian organizations a reason to distinguish “the model knows this passage” from “the system delivered this edition faithfully.” Native generation was especially unreliable for long passages and for the LSV edition. A fluent response should not be assumed to preserve the requested translation.

Three checks follow directly from the results and can be applied to a system an institution is evaluating. Ask whether the requested edition is named and carried through to the delivered text rather than inferred. Ask for evidence that an authorized source was actually consulted for the requested span, since the availability of a lookup tool did not establish its correct use here. And test long passages and event-style requests specifically, because both degraded under every architecture we evaluated.

These results also put a measured figure beside a widely circulated industry estimate that models misquote Scripture between 15% and 60% of the time ([Christian Daily International, 2026](#)). Native generation failed our strict endpoint in 75.00% of observations, which falls inside that range without replicating it: our endpoint demands an exact match to a specified edition on a purposive twenty-passage panel, and the circulated estimate states no protocol. The comparison is offered as calibration, not confirmation.

At the same time, textual fidelity should not be confused with theological fidelity. An exact verse can be detached from its literary context, used to answer the wrong question, or interpreted contrary to a community’s confessional commitments. The present evaluation can help institutions remove misquotation as one failure mode. It cannot decide which passage should be used, how it should be interpreted, or what pastoral counsel should follow. Those tasks require human judgment and, in many settings, accountable ecclesial authority.

6.4 What made requests harder

Two factors changed the character of the task. First, contextual requests required the system to infer a passage before delivering it, and the prompt results show why that extra step deserves separate treatment. When the exact source was already supplied, contextual wording barely changed performance. When the system had to identify and retrieve a passage, contextual descriptions exposed boundary and selection errors. This is not noise around the same task; it is an additional reasoning stage.

Second, the requested edition mattered most when the model generated from memory. Lower native exactness for LSV may reflect source availability, memorization, translation distinctiveness, endpoint behavior, or other factors. The study documents the interaction without treating “familiarity” as a measured training-data variable.

7 Open Evaluation and Reproducibility

The public repository is available at <https://github.com/FideAI/scripture-quotation-fidelity>. It releases:

- this paper and citation metadata;

- the authoritative-quotation methodology and scoring specification;
- scenario and reference schemas;
- the exact executed prompts, functional tool schema, and release-safe target registry;
- the source-edition record, with per-target normalized-text digests for every edition in the locked registry;
- a deidentified 8,640-row derived-score dataset that excludes generated prose and source text;
- reviewed aggregate and target-level tables and a result card;
- the prospective-lock receipt and deviation summary;
- standalone code for the target-level, target-cluster bootstrap, and post-hoc cluster-robust analyses;
- claims, responsible-use, and release policies; and
- documentation mapping the public methodology to a private execution engine.

The public repository excludes private runner code, credentials, provider response identifiers, raw model outputs, held-out prompts, partner-private traces, and restricted source text. This is a deliberate reproducibility boundary. The released derived scores and analysis code reproduce the quantitative claims without network access. An exact endpoint rerun remains subject to model and routing drift and requires an independent execution implementation plus appropriately authorized sources.

The shared condition implementation was executed from a pinned revision of The Apologist Project’s `llm-scripture-fidelity` repository. Fide AI controlled the crossed design, credentials, execution, scoring review, claims boundary, and result custody. The collaboration does not constitute product certification or commercial endorsement. The partner may publish a separate companion explanation of its implementation.

8 Responsible Use and Limitations

This work is a measurement study. It is not legal advice, copyright clearance, theological certification, pastoral-safety approval, product endorsement, or deployment authorization.

Rights status is an architectural input. BSB and WEBU were evaluated as public-domain sources; LSV was used under its stated open license, with no LSV passage text included in the public result package. Restricted translations require separate source and source-to-model authorization. Technical ability to reproduce text does not itself establish a right to distribute it.

For Christian readers, “fidelity” has a deliberately narrow meaning here: fidelity to the requested English source text and span. Christian traditions also use the term for faithfulness to Scripture’s meaning, canonical context, doctrine, and lived obedience. Nothing in the experiment reduces those questions to string matching.

8.1 Limitations of the current study

Purposive target panel. The twenty passages were selected to expose distinct technical behaviors. They do not estimate the prevalence of quotation failure across the Bible, all Christian use cases, or ordinary user traffic.

English and Scripture only. The experiment used three English editions and one sacred-text domain. Reference conventions, translation ecosystems, source authority, and language technology differ elsewhere. Cross-lingual and cross-domain claims require new studies.

Edition versioning. Ground truth was a fixed stored copy of each edition, verified stable by repeat-fetch digests across the run and disclosed in the released source-edition record. Open English editions are nonetheless revised over time, so an exact replication requires the same edition version, not merely the same edition name; the released per-target digests let a replicator confirm that condition before running anything. The digests establish that our ground truth was internally stable and externally checkable, not that the upstream provider’s text is itself authoritative for a given publisher.

Open editions only. The primary experiment does not measure restricted translations. It therefore cannot establish how licensing, provider quotation policies, or source-transfer permissions affect the four architectures.

Model and route drift. Results apply to six named routes on July 26–27, 2026. Hosted models, routing, guardrails, and provider defaults can change. Three epochs measure short-run variability, not longitudinal stability.

Provider-default sampling. Temperature and other generation controls were omitted and left at provider defaults. This improves ecological comparability to the evaluated access path but limits causal interpretation of sampling parameters.

Strict interface contract. The deterministic primary result includes failures caused by a deliberately strict reference grammar. Tolerant analysis shows that many syntax-tagged outputs contained the intended reference. A production system could adopt a more permissive parser, but changing the parser after observing results would invalidate the locked primary comparison.

Automated exactness. Exact matching is appropriate for the declared endpoint, but failure tags and normalized views depend on implementation choices. A post-run route-blinded review checked mechanisms in a small diagnostic sample; it was not a second population annotation study.

Dependence and fixed-panel inference. Target-cluster bootstrap intervals describe variation within this target panel, while model routes and editions remain fixed. Target-level distributions and a post-hoc target-cluster-robust fixed-panel sensitivity analysis make this dependence visible, but neither supports population inference to all Scripture requests, models, or editions. With only twenty target clusters, uncertainty estimates remain coarse; a larger probability-oriented target sample would be needed for population claims.

Contextual-description review. The six revised contextual descriptions received a structurally blinded technical reapproval, but the reviewer was an AI system and the same reviewing identity that authored the initial critique. This supports an internal construct check, not independent credentialed scholarly validation. The contextual-prompt results should be read with that limitation attached: a low reference-identification rate could reflect system behavior or description quality, and the present evidence does not fully separate them. Materials for an independent scholarly review of all twenty descriptions—protocol, blinded packet, and pre-committed scoring rules—are

released with this package so that check can be carried out and reported by us or by others. Primary scores are frozen, so such a review can qualify how the contextual-prompt findings are interpreted but cannot alter any reported quantity.

No contextual or theological endpoint. Contextual descriptions test technical reference identification. The study does not determine whether the expected passage is the best theological, exegetical, or pastoral response to a real person’s situation.

9 Conclusion

AI systems should not always generate. In this study, native parametric quotation delivered the requested edition exactly in 25.00% of observations. Supplying authoritative text, connecting a source tool, or delegating final rendering to a deterministic layer all produced large improvements, but none eliminated failure. Each architecture moved failure to the decision it still left to the model.

The practical lesson is not that one method wins every comparison. It is that authoritative quotation should be designed and evaluated as a sequence of observable responsibilities. Models may identify what a user means. Source interfaces may retrieve what is authorized. Deterministic components may render what must not change. Final-output checks must establish what the user actually received.

Return to the opening request about Jesus and the serpent in the wilderness. The system should be able to show whether the model identified the intended passage, whether an authorized source supplied the requested edition, and whether the final words survived intact. If the answer is wrong, the same chain should reveal whether the error was selection, access, or rendering.

Scripture makes these distinctions visible, but the underlying problem belongs to any AI system that handles authoritative text. When wording is material, the system should be able to say when the model is speaking and when the source of record is speaking.

Data and Code Availability

The public repository, <https://github.com/FideAI/scripture-quotation-fidelity>, releases the full deidentified derived-score dataset, target registry, executed prompt templates, functional tool schema, source-edition record with per-target text digests, release-safe prospective-lock and deviation records, and standalone analysis code. It excludes generated prose, authoritative passage text, provider response identifiers, raw tool traces, credentials, restricted-source payloads, and private execution-engine code. The released package reproduces the quantitative tables and contrasts without model endpoint access.

Author Contributions

Alex Chao: conceptualization, methodology, study design, software integration, validation, formal analysis, investigation, data curation, visualization, writing, supervision, and project administration.

Competing Interests and Collaboration

The Apologist Project collaborated on and maintains the shared condition implementation used for local execution. Fide AI controlled the methodology, prospective lock, credentials, confirmatory execution, analysis, evidence custody, release decision, and claims boundary. The Apologist Project develops commercial technology related to Scripture delivery, creating an organizational interest relevant to the study topic. The study did not evaluate or certify a deployed commercial product.

Use of AI Systems

AI systems assisted with code generation, manuscript drafting, citation cross-checking, and internal contextual-description review. Quantitative outcomes were produced by the declared automated scoring pipeline. The AI review is disclosed as an internal construct check and is not represented as independent credentialed scholarly validation.

A Target References

The twenty target references were: John 3:16; Genesis 1:1; Romans 8:28; Hebrews 11:1; John 14:6; Psalm 23:1–3; Proverbs 3:5–6; Ephesians 2:8–9; Philippians 4:6–7; Deuteronomy 6:4–5; Matthew 5:3–10; 1 Corinthians 13:4–7; Isaiah 53:4–6; Revelation 21:1–4; Matthew 22:37–40; Numbers 21:8–9; John 3:14–15; Luke 2:10–12; Micah 6:8; and James 1:5.

The public repository includes all twenty release-safe target records and the exact contextual descriptions. Generated outputs and authoritative source text remain governed by the release policy.

B Execution Routes and Prospective Lock

The executed OpenRouter model identifiers and pinned upstreams were: `openai/gpt-5.6-sol` via OpenAI; `anthropic/claude-sonnet-5` via Anthropic; `google/gemini-3.5-flash` via Google AI Studio; `deepseek/deepseek-v4-pro` via Together; `moonshotai/kimi-k3` via Moonshot; and `z-ai/glm-5.2` via Together. Fallbacks were disabled, required parameter support was enabled, and data-collecting routes were denied.

The internal prospective lock was recorded at 2026-07-26T22:28:40Z. Confirmatory execution began at 2026-07-26T23:11:30Z. The lock identified the target pack, execution matrix, contextual annotations, source panel and registry, scoring rubric, manifest, and deviation log by SHA-256 digest. These records are intended for the archival replication package.

C Route-by-Condition Results

D Failure Tags

Observed structured tags included 345 wrong or unparseable deterministic reference selections, 234 tool calls that did not retrieve the requested span, 161 invalid deterministic reference payloads or placeholder failures, 161 selected-reference lookup failures, and 107 tool-bypass observations. Tags can co-occur and must not be summed as unique failed observations. Native and source-supplied text mismatches are represented primarily through exact, normalized, coverage, and answer-presence metrics rather than these architecture-specific tags.

Evaluated route	Native	Source supplied	Tool	Deterministic
Claude Sonnet 5	12.78%	82.22%	78.06%	75.56%
DeepSeek V4 Pro	32.22%	96.11%	83.61%	89.17%
Gemini 3.5 Flash	16.11%	98.06%	86.94%	91.94%
GLM-5.2	20.00%	93.61%	75.56%	85.83%
GPT-5.6 Sol	35.28%	97.22%	92.50%	71.39%
Kimi K3	33.61%	94.44%	63.89%	89.72%

Table 7: Descriptive route-by-condition architecture-adherent exactness ($N = 360$ per cell). Rows are ordered alphabetically, not by performance; this is not a model leaderboard.

E Claims Boundary

Results apply only to the named routes, dates, twenty-target panel, two prompt families, three open English editions, four architecture conditions, and declared scoring protocol. The study does not establish:

- a general ranking of models or vendors;
- theological truth or contextual appropriateness;
- pastoral safety or institutional deployment readiness;
- training-data membership or memorization causation;
- universal legal or licensing compliance;
- performance on restricted translations; or
- superiority of a commercial product.

References

- Barna Group. 2026. New research: Pastors are using AI more than you think. <https://www.barna.com/research/pastors-using-ai-ministry/>. Published June 15, 2026. Online surveys of U.S. Protestant pastors ($n = 442$, December 2025) and U.S. adults ($n = 1,514$, November 2025); accessed 2026-07-29.
- Berean Bible. 2023. Free licensing. <https://berean.bible/licensing.htm>. Licensing statement; accessed 2026-07-27.
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. 2023. [Quantifying memorization across neural language models](#). In *The Eleventh International Conference on Learning Representations*. OpenReview.net.
- Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. 2021. [Extracting training data from large language models](#). In *30th USENIX Security Symposium (USENIX Security 21)*, pages 2633–2650. USENIX Association.
- Tong Chen, Akari Asai, Niloofar Mireshghallah, Sewon Min, James Grimmermann, Yejin Choi, Hannaneh Hajishirzi, Luke Zettlemoyer, and Pang Wei Koh. 2024. [CopyBench: Measuring literal and non-literal reproduction of copyright-protected text in language model generation](#). In

- Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15134–15158, Miami, Florida, USA. Association for Computational Linguistics.
- Christian Daily International. 2026. AI’s bible misquotes range from 15% to 60%, YouVersion CEO says. <https://www.christianpost.com/news/ais-bible-misquotes-range-from-15-to-60-youversion-ceo.html>. Published March 17, 2026 in The Christian Post. Reported industry estimate; no measurement methodology given; accessed 2026-07-29.
- Covenant Press. 2020. Literal standard version. <https://www.bible.com/en-GB/versions/2660-lsv-literal-standard-version>. Edition and CC BY-SA licensing statement; accessed 2026-07-27.
- eBible.org. 2024. The world english bible (WEB) faq. <https://ebible.org/eng-web/webfaq.htm>. Updated February 5, 2024; accessed 2026-07-27.
- Shahul Es, Jithin James, Luis Espinosa Anke, and Steven Schockaert. 2024. [RAGAs: Automated evaluation of retrieval augmented generation](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158, St. Julians, Malta. Association for Computational Linguistics.
- Exponential AI NEXT and ChurchTechToday. 2025. The 2025 state of AI in the church national survey report. <https://churchtechtoday.com/2025-state-of-ai-in-the-church/>. Survey of 594 church leaders; accessed 2026-07-29.
- Fide AI. 2026. FID-056: Formal verification for sacred text fidelity. <https://github.com/FideAI/research-ideas/blob/main/ideas/FID-056-formal-verification-sacred-text-fidelity.md>. Public research call; published at <https://fideai.org/research/calls>; accessed 2026-07-30.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. [Enabling large language models to generate text with citations](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488, Singapore. Association for Computational Linguistics. Introduces the ALCE benchmark.
- Saibo Geng, Martin Josifoski, Maxime Peyrard, and Robert West. 2023. [Grammar-constrained decoding for structured NLP tasks without finetuning](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 10932–10952, Singapore. Association for Computational Linguistics.
- Philip Goff, Arthur E. Farnsley, II, and Peter J. Thuesen, editors. 2017. *The Bible in American Life*. Oxford University Press, New York. Includes General Social Survey and National Congregations Study modules on which translation respondents read.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2025. [A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions](#). *ACM Transactions on Information Systems*, 43(2):1–55.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. [Survey of hallucination in natural language generation](#). *ACM Computing Surveys*, 55(12):1–38.

- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive nlp tasks](#). *Advances in Neural Information Processing Systems*, 33:9459–9474.
- Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christopher Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, and 31 others. 2023. [Holistic evaluation of language models](#). *Transactions on Machine Learning Research*.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. [When not to trust language models: Investigating effectiveness of parametric and non-parametric memories](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822, Toronto, Canada. Association for Computational Linguistics.
- Bruce M. Metzger and Bart D. Ehrman. 2005. *The Text of the New Testament: Its Transmission, Corruption, and Restoration*, 4 edition. Oxford University Press, New York.
- Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. [FActScore: Fine-grained atomic evaluation of factual precision in long form text generation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore. Association for Computational Linguistics.
- Eugene A. Nida and Charles R. Taber. 1969. *The Theory and Practice of Translation*, volume 8 of *Helps for Translators*. E. J. Brill, Leiden.
- Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. [Beyond accuracy: Behavioral testing of NLP models with CheckList](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912, Online. Association for Computational Linguistics.
- Hayley Ross, Ameya Sunil Mahabaleshwarkar, and Yoshi Suhara. 2025. [When2Call: When \(not\) to call tools](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3391–3409, Albuquerque, New Mexico. Association for Computational Linguistics.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. [Toolformer: Language models can teach themselves to use tools](#). In *Advances in Neural Information Processing Systems*, volume 36.
- Emanuel Tov. 2012. *Textual Criticism of the Hebrew Bible*, 3, revised and expanded edition. Fortress Press, Minneapolis.
- United States Copyright Office. 2025. Copyright and artificial intelligence, part 3: Generative ai training (pre-publication version). <https://www.copyright.gov/ai/Copyright-and-Artificial-Intelligence-Part-3-Generative-AI-Training-Report-Pre-Publication-Version.pdf>. Report of the Register of Copyrights, released May 2025.

Jingyu Zhang, Marc Marone, Tianjian Li, Benjamin Van Durme, and Daniel Khashabi. 2025. [Verifiable by design: Aligning language models to quote from pre-training data](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3748–3768, Albuquerque, New Mexico. Association for Computational Linguistics.